

1 Life-Cycle Model With Endogenous Labor Supply

- State variables:
 - a , assets.
 - η , transitory labor productivity shock.
 - j , age.
 - θ , permanent productivity type.
- Choice variables:
 - a' , next-period assets.
 - l , labor supply, or hours of work.
- Household optimization problem in recursive form:

$$V(a, \eta, j, \theta) = \max_{a', l} \left\{ u(c, 1 - l) + \beta \psi_{j+1} E [V(a', \eta', j + 1, \theta) \mid \eta] \right\}.$$

The budget constraint is

$$c + a' = (1 + r)a + w e_j \theta \eta l + pen_j,$$

with

$$a' \geq 0, \quad 0 \leq l \leq 1, \quad c \geq 0.$$

Given a' , labor supply has the following closed-form expression. Define

$$available = (1 + r)a + pen_j, \quad wage = w e_j \theta \eta.$$

For working ages, $j < J_R$,

$$\tilde{l}(a', a, \eta, j, \theta) = \nu + (1 - \nu) \frac{a' - available}{wage},$$

and the implemented labor supply is

$$l(a', a, \eta, j, \theta) = \min \left\{ \max \left\{ \tilde{l}(a', a, \eta, j, \theta), 0 \right\}, 1 - 10^{-10} \right\}.$$

For retirement ages, $j \geq J_R$,

$$l(a', a, \eta, j, \theta) = 0.$$

The policy functions are

$$\begin{aligned} a' &= g_a(a, \eta, j, \theta), \\ l &= g_l(a, \eta, j, \theta). \end{aligned}$$

Effective labor productivity is

$$h_j(\eta, \theta) = \begin{cases} e_j \theta \eta, & j < J_R, \\ 0, & j \geq J_R, \end{cases}$$

where e_j is the deterministic age-efficiency profile. In the Fortran code, $r = 0.04$, $w = 1$, $J = 80$, $J_R = 45$, $N_a = 200$, $N_\theta = 2$, and $N_\eta = 7$. The pension is

$$pen_j = \begin{cases} 0, & j < J_R, \\ \kappa \tau_p \frac{1}{J_R - 1} \sum_{s=1}^{J_R-1} e_s, & j \geq J_R, \end{cases}$$

where $\kappa = 0.5$ is the replacement parameter and $\tau_p = 0.33$ is the pension contribution parameter. The period utility function is

$$u(c, 1 - l) = \frac{[c^\nu (1 - l)^{1-\nu}]^{1-\frac{1}{\gamma}}}{1 - \frac{1}{\gamma}},$$

with $\nu = 0.335$, $\gamma = 0.50$, and $\beta = 0.98$.

The transitory productivity process is discretized as an AR(1) process,

$$z' = \rho z + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon),$$

with $\rho = 0.985$ and innovation variance $\sigma_\varepsilon = 0.022$. The corresponding innovation standard deviation is $\sqrt{0.022}$. The code stores $\eta = \exp(z)$ after discretization. The two permanent productivity types are

$$\theta_1 = \exp(-\sqrt{\sigma_\theta}), \quad \theta_2 = \exp(\sqrt{\sigma_\theta}),$$

with $\sigma_\theta = 0.242$ and equal type probabilities.

Initial condition: at age 1, all individuals start with zero assets and the middle transitory shock state, which corresponds to $z = 0$ and hence $\eta = 1$. The permanent productivity type is distributed uniformly across the two permanent types.

- Distribution:

$$\mu(a', \eta', j+1, \theta) = \sum_a \sum_{\eta} I\{a' = g_a(a, \eta, j, \theta)\} \pi(\eta, \eta') \mu(a, \eta, j, \theta),$$

for $j = 1, \dots, J-1$. If $g_a(a, \eta, j, \theta)$ is not a grid point, the code linearly redistributes mass across the two neighboring asset-grid points. Let $a_k \leq a' < a_{k+1}$ and choose ω such that

$$a' = \omega a_k + (1 - \omega) a_{k+1}.$$

Then the algorithm assigns the fraction ω of the mass to a_k and the fraction $1 - \omega$ to a_{k+1} .

The initial distribution is

$$\mu(a, \eta, 1, \theta_i) = \begin{cases} \pi_i^\theta, & a = 0, \eta = 1, \theta = \theta_i, \\ 0, & \text{otherwise.} \end{cases}$$

2 Mapping Into VFI-Toolkit Notation

In VFI-Toolkit notation, θ is a permanent type and should be handled through the permanent-type dimension. The shock η is the exogenous Markov shock, corresponding to z in the toolkit notation. Assets a are the endogenous state, while a' is the next-period asset choice.

Labor l is conceptually a decision variable, so it corresponds to d in the toolkit notation. However, because the closed-form solution gives l as a function of a' , a , η , j , and θ , we do not need to declare a separate d grid. The return function can instead compute labor internally and use (a', a, η, j, θ) as the relevant action-state inputs, with age j handled through the finite-horizon setup.